Module 5

Options – Basics

September 24 & 29, 2025

Contract:

- Call: right but not obligation to buy asset S at a strike price K in some future dates.

  Payoff = max(S – K, 0)

- Put: right but not obligation to sell asset S at a strike price K in some future dates.

  Payoff = max(K – S, 0)

- European style: option can only be exercised at its expiry (maturity) T
- American style: option can be exercised at anytime before expiry $0 < t \leq T$

Options with such payoffs are called vanilla options

Terminology:

- Intrinsic Value (IV): present value of exercising option at maturity T with current stock price

  $IV = \max(S - Ke^{-rT}, 0)$ call

  $IV = \max(Ke^{-rT} - S, 0)$ put

- Moneyness:
  - ITM (IV > 0) (In-The-Money)
  - ATM ($S = Ke^{-rT}$) (At-The-Money)
  - OTM ($S < Ke^{-rT}$ for call, $S > Ke^{-rT}$ for put) (Out-The-Money)
- Time Value (TV) = Option Premium − Intrinsic Value (IV)

Note:

Some people may refer IV, ITM, ATM, OTM using slightly different conventions. For instance for call options IV is measured by $S - K$ or $Se^{rT} - K$

Option positions:

- Long call
- Long put
- Short (write) call (margin account required)
- Short (write) put (margin account required)

Underlying asset for options:

- stock, stock indices, ETF, bonds, bond indices, …
- FX
- Futures
- commodity
- …

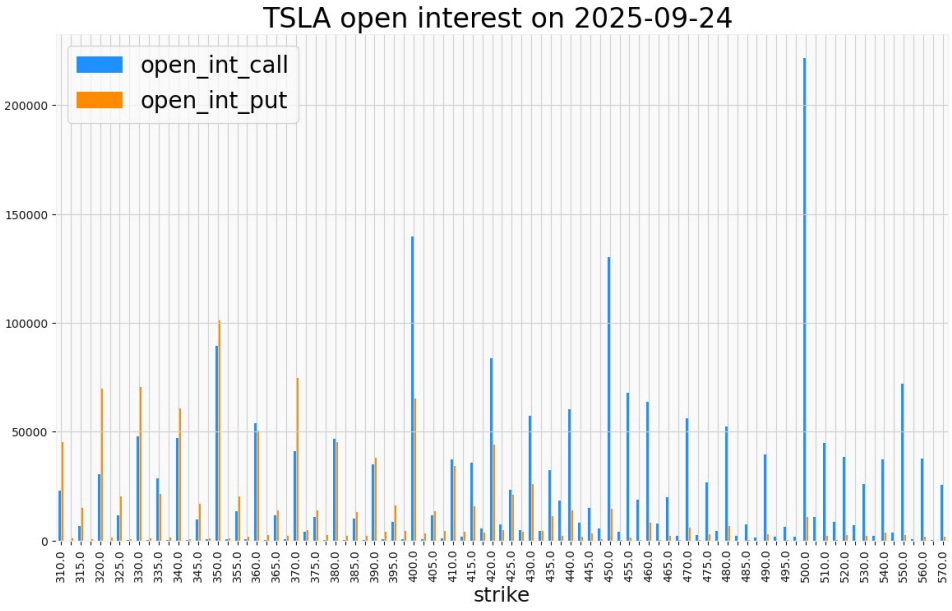
Option Exchanges:

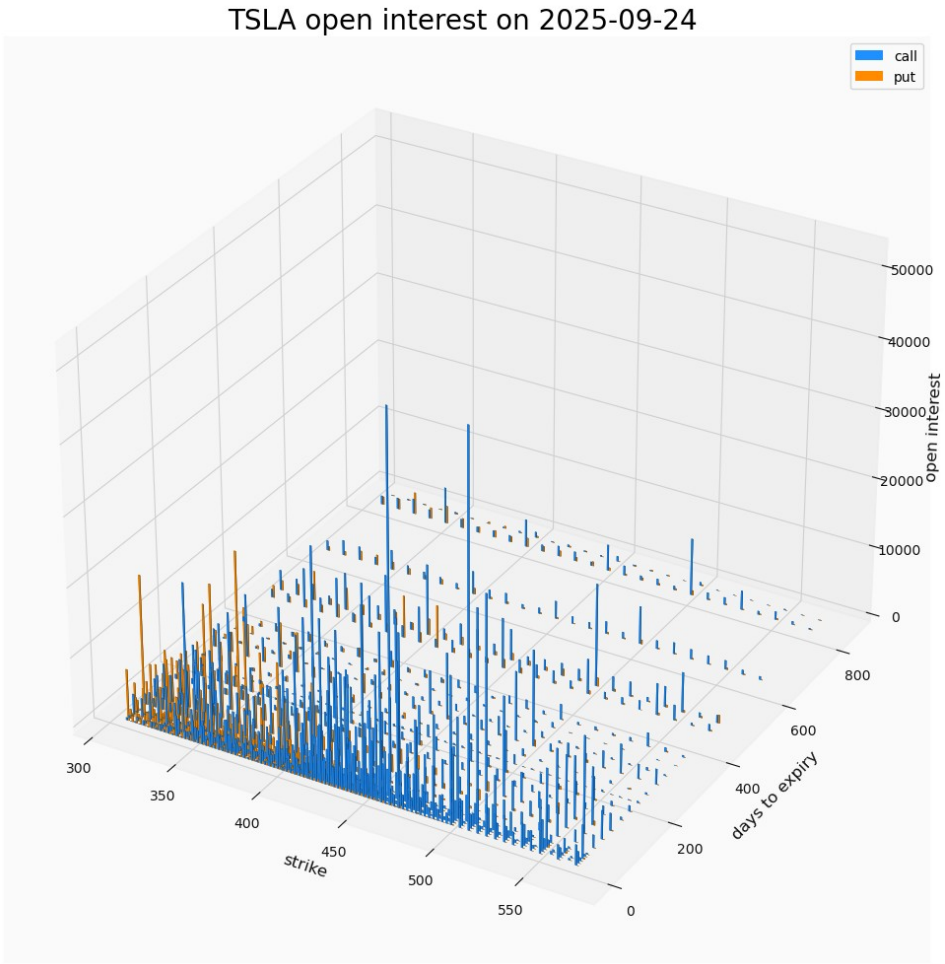
- Specifications in exchanges: expirations, contract sizes, strikes

  e.g., for equity, 1 contract = 100 shares
- Most exchange traded options are American style
- No dividend protection in general
- LEAPS: long dated options (time to expiry greater than 1 year)
- FLEX options: exchange traded options on stock/stock index with non-standard expiries and strikes (mostly short dated options on super liquid assets)
- 0DTE options: zero days to expiration options
- Position limits and exercise limits
- Margin requirement for option writers

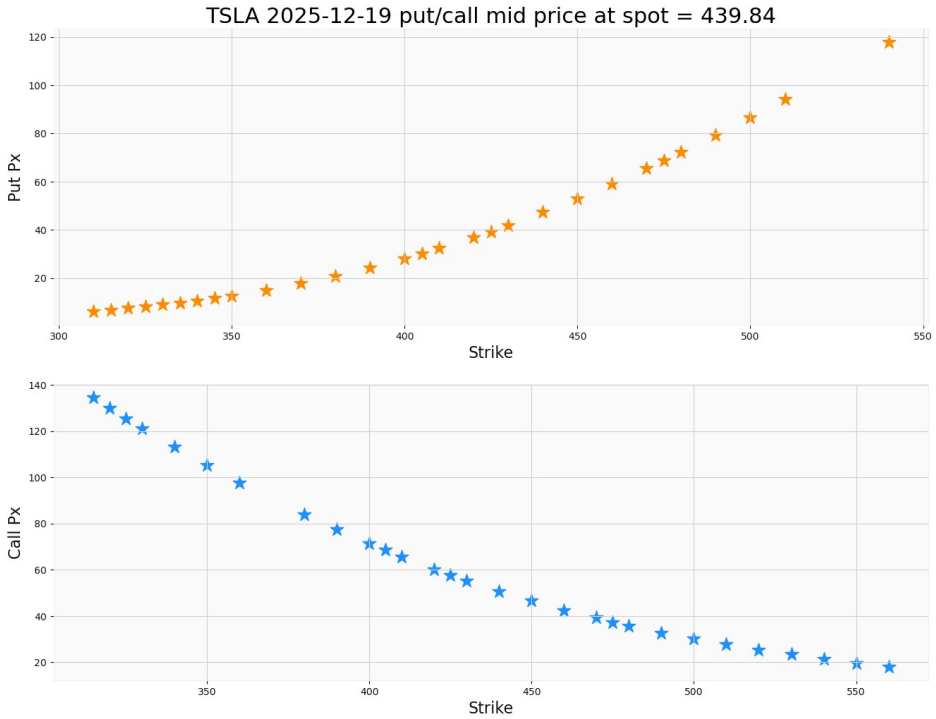
Example:

TSLA option quote snapshot on September 24, 2025 (source: Yahoo Finance):


TSLA open interest on 2025-09-24
open_int_call
open_int_put
200000
150000
100000
50000
0
310.0 315.0 320.0 325.0 330.0 335.0 340.0 345.0 350.0 355.0 360.0 365.0 370.0 375.0 380.0 385.0 390.0 395.0 400.0 405.0 410.0 415.0 420.0 425.0 430.0 435.0 440.0 445.0 450.0 455.0 460.0 465.0 470.0 475.0 480.0 485.0 490.0 495.0 500.0 510.0 520.0 530.0 540.0 550.0 560.0 570.0
strike


TSLA open interest on 2025-09-24
call
put
50000
40000
30000
20000
10000
0
open interest
300
350
400
450
500
550
strike
0
200
400
600
800
days to expiry

Option prices sampled on 2025-09-24





Consider a call option with strike K = 444.18, expiration = December 19, 2025 (3rd Friday of the month). At S_0 = 439.84 it is traded at 53.07 (source: Yahoo Finance, September 24, 2025, interpolated mid price)

Compare payoffs at expiry with outright stock ownership, forward and TRS (single exchange):

As of date	9/24/2025

r	4.25%
T	0.2320
S0 (TSLA)	439.84

Option Strike	444.18
Call Px	53.07
Contract	0.01
expiry	12/19/2025

Mkt Move\Payoff	Outright	Fwd	TRS	Call
-50.0%	219.92	-224.26	-224.26	0.00
-20.0%	351.87	-92.30	-92.30	0.00
-10.0%	395.86	-48.32	-48.32	0.00
-5.0%	417.85	-26.33	-26.33	0.00
-2.5%	428.84	-15.33	-15.33	0.00
0.0%	439.84	-4.34	-4.34	0.00
2.5%	450.84	6.66	6.66	6.66
5.0%	461.83	17.66	17.66	17.66
10.0%	483.82	39.65	39.65	39.65
20.0%	527.81	83.63	83.63	83.63
50.0%	659.76	215.58	215.58	215.58

Forward Px	444.18
Contract size	1.00

TRS Notional	439.84

Mkt Move\% PnL	Outright	Fwd	TRS	Call
-50.0%	-50.0%	NA	NA	-100.0%
-20.0%	-20.0%	NA	NA	-100.0%
-10.0%	-10.0%	NA	NA	-100.0%
-5.0%	-5.0%	NA	NA	-100.0%
-2.5%	-2.5%	NA	NA	-100.0%
0.0%	0.0%	NA	NA	-100.0%
2.5%	2.5%	NA	NA	-87.5%
5.0%	5.0%	NA	NA	-66.7%
10.0%	10.0%	NA	NA	-25.3%
20.0%	20.0%	NA	NA	57.6%
50.0%	50.0%	NA	NA	306.3%

→ It is nice to have an option, but it comes with a price!

Examples of option usage:

- Long call/put for hedging:
    - a pension fund long a call on an index when it expects cash inflow
    - a pension fund long a put on an index when it expects cash outflow while participate upside
    - long deep OTM puts to hedge market crash (deep OTM options are cheap)

- Long call/put for speculations
- Writing covered calls
- Writing puts to acquire stock positions
- Financial Engineering: PPN (Principal Protected Notes)

Factors affecting option prices:

- S
- K
- T
- r
- dividend
- σ : stock return volatility

The first 5 factors we already encountered while valuing forwards, the last one is new:

- Payoff of a long forward = S – K, "linear", S is in the entire payoff spectrum
- Payoff of a call = max(S – K, 0), "non-linear", need to somehow know how likely S will end up ITM

General properties:

Let C, P = American call, put; c, p = European call, put; present time t = 0

- Options have monetary value:

  C, P, c, p $\geq$ 0

- Option value increases with optionality:
  - If K, T are the same for C and c, or P and p $\rightarrow$ C $\geq$ c, P $\geq$ p
  - T $\nearrow$ $\rightarrow$ C $\nearrow$, P $\nearrow$
- Option value increases with moneyness:
  - K $\nearrow$ $\rightarrow$ P $\nearrow$, p $\nearrow$
  - K $\searrow$ $\rightarrow$ C $\nearrow$, c $\nearrow$

Bounds on option value:

For simplicity and to focus on general idea we assume no dividend payments in the discussion below unless explicitly indicated:

Upper bounds on options:

- Call options are bounded by the underlying asset prices:

  $c \leq S$, $C \leq S$

  if not: short call, use the proceed to long share → arbitrage

- Put options are bounded by their strikes:

  $p \leq Ke^{-rT}$, $P \leq K$

  if not: short put, let proceed to grow risk free → arbitrage

Lower bounds on options:

Options are bounded from below by their intrinsic values:

- $c \geq S - Ke^{-rT}$

  If not (assume $S > Ke^{-rT}$, i.e., $c > 0$): short stock, use the proceed to long call, deposit the difference A ($> Ke^{-rT}$);

  at T, cash grows to $A_T > K$.

  - if $S_T > K$, exercise call using K, cover the short with the share received from call, cash remaining $A_T - K > 0$
  - if $S_T \leq K$, let call expire, cover short using cash = $S_T < K$, fund remaining $A_T - S_T > 0$

  → arbitrage

- $p \geq Ke^{-rT} - S$

  If not (assume $S < Ke^{-rT}$, i.e., $p > 0$): borrow S to long stock, borrow p to long put, total debt = S + p = B ($< Ke^{-rT}$);

  at T, debt matures at $B_T < K$.

  - if $S_T \leq K$, exercise put using S to receive K, pay back B_T , cash remaining $K - B_T > 0$
  - if $S_T > K$, let put expire, sell stock to receive S_T , pay back B_T , remaining fund = $S_T - B_T > 0$

  → arbitrage

- $C \geq \max(S - K, 0)$, $P \geq \max(K - S, 0)$
    - If not, long option and exercise immediately → arbitrage

Summary:

$\max(S - Ke^{-rT}, 0) \leq c \leq S$, $\max(Ke^{-rT} - S, 0) \leq p \leq Ke^{-rT}$

$\max(S - Ke^{-rT}, 0) \leq c \leq C \leq S$, $\max(Ke^{-rT} - S, 0) \leq p \leq P \leq K$

Time Value:

- C should never be exercised early (if div = 0, r > 0):

  - if exercised: $C = S - K \leq S - Ke^{-rT}$

  - if not exercised: $C \geq c \geq S - Ke^{-rT}$

- P may be exercised early even if div = 0:

  - if exercised: $P = K - S$

  - if not exercised: $P = IV + TV = Ke^{-rT} - S + TV$

  exercise only if $K - S > P$ (exercise boundary) :

  $K - S > Ke^{-rT} - S + TV$ → $K(1 - e^{-rT}) > TV$

  $K(1 - e^{-rT}) = PV(K(e^{rT} - 1))$ = present value of interest in K

  → exercise if interest on K from early exercise > option time value

Effect of dividend:

Assume ex-dividend date = t^+, $0 < t^+ < T$, with dividend D_t

- early exercise C may be optimal:

  if exercised at t (just prior t^+): $C = S_t - K = S_{t^+} + D_t - K$ (received dividend since owning S now)

  if not exercised, $C = S_{t^+} - Ke^{-r(T-t)} + TV$

  exercise if $S_{t^+} + D_t - K > S_{t^+} - Ke^{-r(T-t)} + TV$, or

  $D_t > K(1 - e^{-r(T-t)}) + TV$

→ exercise if D_t can make up the interest lost from K and lost of time value

→ the larger D_t , the likely early exercised.

- early exercise P:

  at t = 0, S has PV(D)

  if exercised: $P = K - S$

  if not exercised: $P = Ke^{-rT} - (S - PV(D)) + TV$

  exercise if $K - S > Ke^{-rT} - (S - PV(D)) + TV$, or

  $K(1 - e^{-rT}) > PV(D) + TV$

  → exercise if interest on K from early exercise > lost dividend + option time value (put is hedged → long S)

  → the larger the dividend amount paid during the life of put, the less likely it gets exercised

Remark: condition for optimal early exercise will be made precise in advanced courses

<u>Put-call parity</u>:

- European options:

  long call + short put = long forward (with the same K, T)

  Consider a stock pays no dividend, let t = 0:

  $c - p = S - Ke^{-rT}$ → $c + Ke^{-rT} = p + S$

  Recall the inequality we established earlier:

  $c \geq S - Ke^{-rT}$

  $p \geq Ke^{-rT} - S$

  Compare with the put-call parity we see that the differences are precisely given by the put and call price respectively.

  Since the statement "long call + short put = long forward" is valid in general regardless how dividend is paid, we can immediately write put-call parity for

  - stock pays discrete dividend payments with PV(D)

$c - p = S - PV(D) - K\,e^{-rT} \rightarrow c + K\,e^{-rT} = p + S - PV(D)$

- stock with continuous dividend yield q

$c - p = Se^{-qT} - Ke^{-rT} \rightarrow c + Ke^{-rT} = p + Se^{-qT}$

- American options:

Stock pays no dividend:

$S - K \le C - P \le S - Ke^{-rT}$

Proof:

(1) $P \ge p = c + Ke^{-rT} - S$, $C = c$

$\rightarrow P \ge C + Ke^{-rT} - S$

$\rightarrow C - P \le S - Ke^{-rT}$

(2) Consider portfolio H consisted of one long C, one short P, one short S, K amount cash:

$H = C - P - S + K$

If P exercised early at $t < T$:

$H_t = C_t - (K - S_t) - S_t + Ke^{rt} = C_t + K(e^{rt} - 1) \ge 0$

If P not exercised early, then at T

if $S_T \le K$

$H_T = 0 - (K - S_T) - S_T + Ke^{rT} = K(e^{rT} - 1) \ge 0$

if $S_T > K$

$H_T = (S_T - K) - 0 - S_T + Ke^{rT} = K(e^{rT} - 1) \ge 0$

$H_T \ge 0 \ \forall \ S_T \rightarrow H_t \ge 0 \ \forall \ S_t$

$\rightarrow H_t \ge 0$ regardless whether P is exercised early

$\rightarrow C - P - S + K \ge 0$

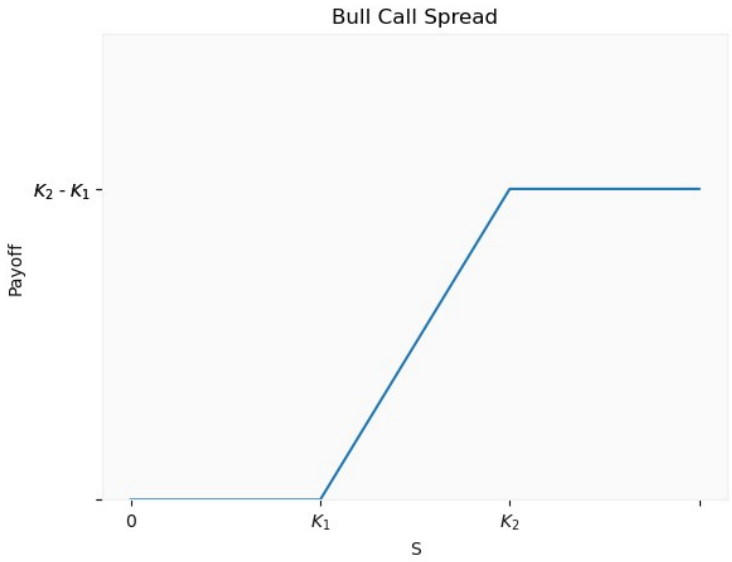
$\rightarrow S - K \le C - P \le S - Ke^{-rT}$

Examples of option strategies:

Assume European style options:

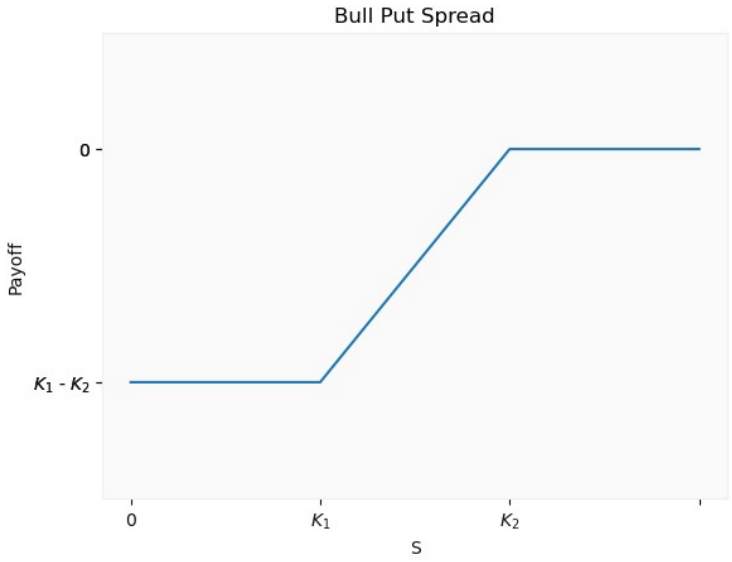
- Bull/bear spreads – same T, different K: let $K_1 < K_2$
    - Bull spread with calls: short K_2 call, long K_1 call

$$\text{Payoff} = \max(S - K_1, 0) - \max(S - K_2, 0) = \begin{cases} K_2 - K_1 & K_2 \le S \\ S - K_1 & K_1 < S < K_2 \\ 0 & S \le K_1 \end{cases}$$



    - Bull spread with puts: short K_2 put, long K_1 put

$$\text{Payoff} = \max(K_1 - S, 0) - \max(K_2 - S, 0) = \begin{cases} 0 & K_2 \le S \\ S - K_2 & K_1 < S < K_2 \\ K_1 - K_2 & S \le K_1 \end{cases}$$



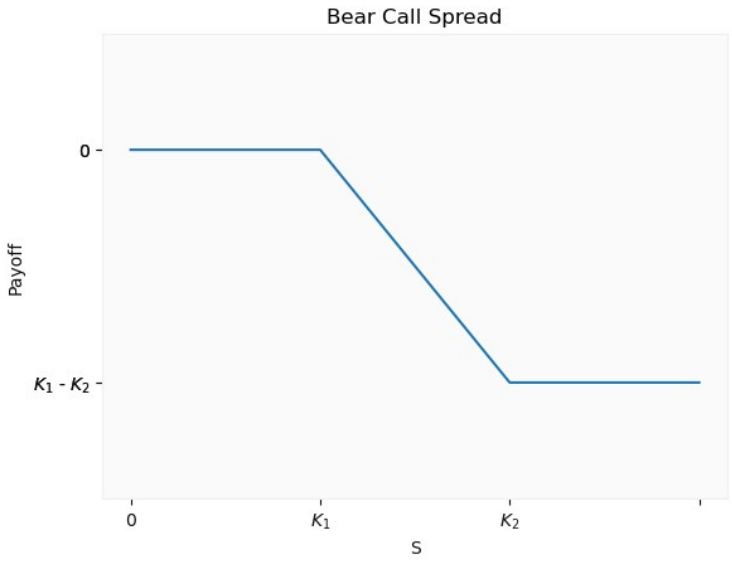
Why does one want to do that with puts?

- bull spread with puts → receive up front premium

- bull spread with calls → pay up front cost

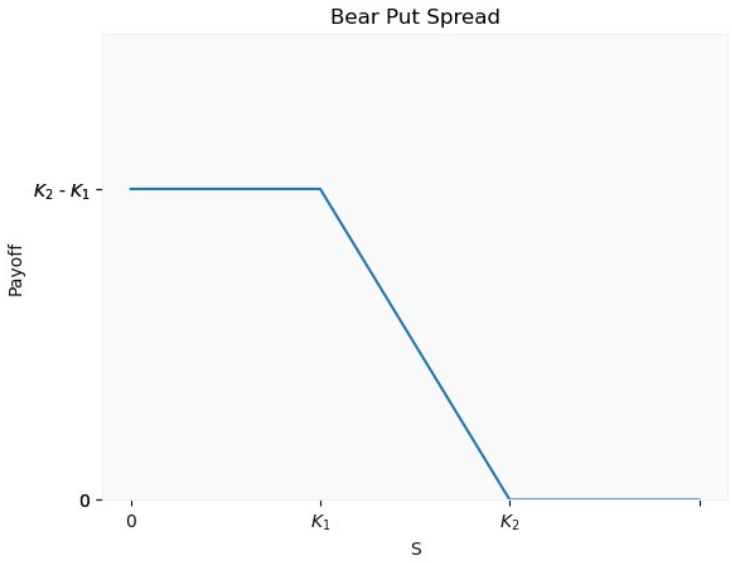
- Bear spread with calls: short K_1 call, long K_2 call

$$\text{Payoff} = \max(S - K_2, 0) - \max(S - K_1, 0) = \begin{cases} K_1 - K_2 & K_2 \le S \\ K_1 - S & K_1 < S < K_2 \\ 0 & S \le K_1 \end{cases}$$



- Bear spread with puts: short K_1 put, long K_2 put

$$\text{Payoff} = \max(K_2 - S, 0) - \max(K_1 - S, 0) = \begin{cases} 0 & K_2 \le S \\ K_2 - S & K_1 < S < K_2 \\ K_2 - K_1 & S \le K_1 \end{cases}$$



- bear spread with calls → receive up front premium

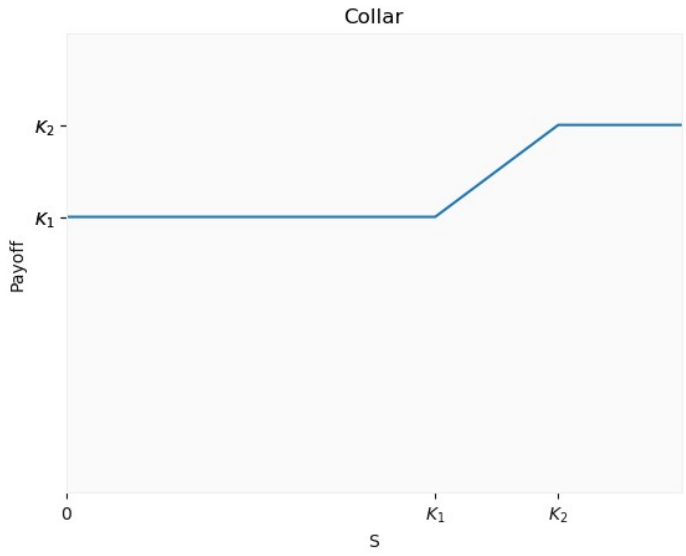
- bear spread with puts → pay up front cost

- Collar (may be used to protect a long position)

  Long share S, short call at K_2, long put at $K_1 < K_2$, both options matures at T

  → share price is locked in $[K_1, K_2]$ at T

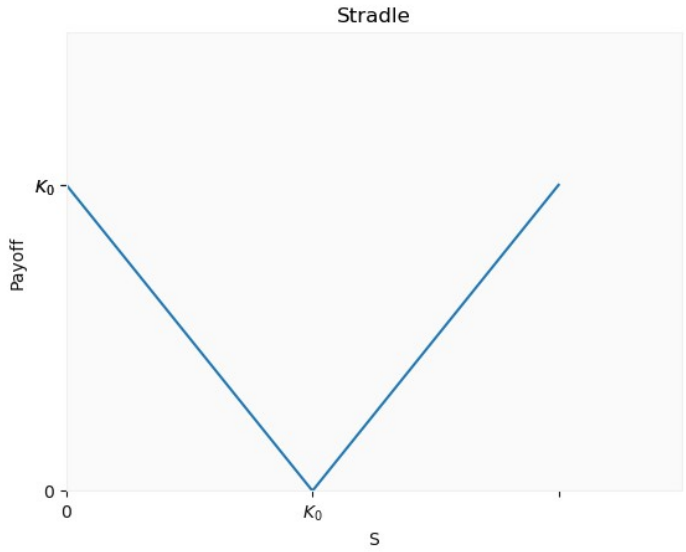
  When price of the K_2 call = K_1 put → "zero cost collar"



- Straddles – same T, same K = K_0

  Long straddle: long call at K_0, long put at K_0 (often K_0 is selected such that it is at or close to S_0)

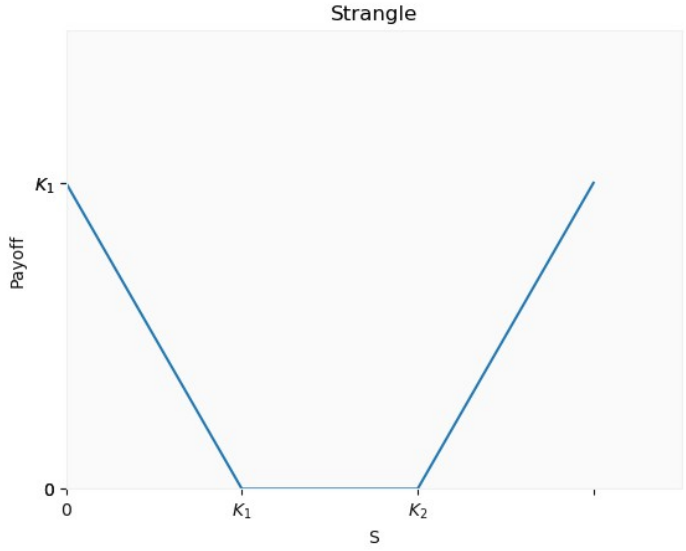
  Payoff = $\max(S - K_0, 0) + \max(K_0 - S, 0) = \max(S - K_0, K_0 - S)$



- Strangles – same T, different K: let $K_1 < K_2$

  Long strangle: long call at K_2, long put at K_1

  Payoff = $\max(K_1 - S, 0) + \max(S - K_2, 0) = \begin{cases} S - K_2 & K_2 \le S \\ 0 & K_1 < S < K_2 \\ K_1 - S & S \le K_1 \end{cases}$

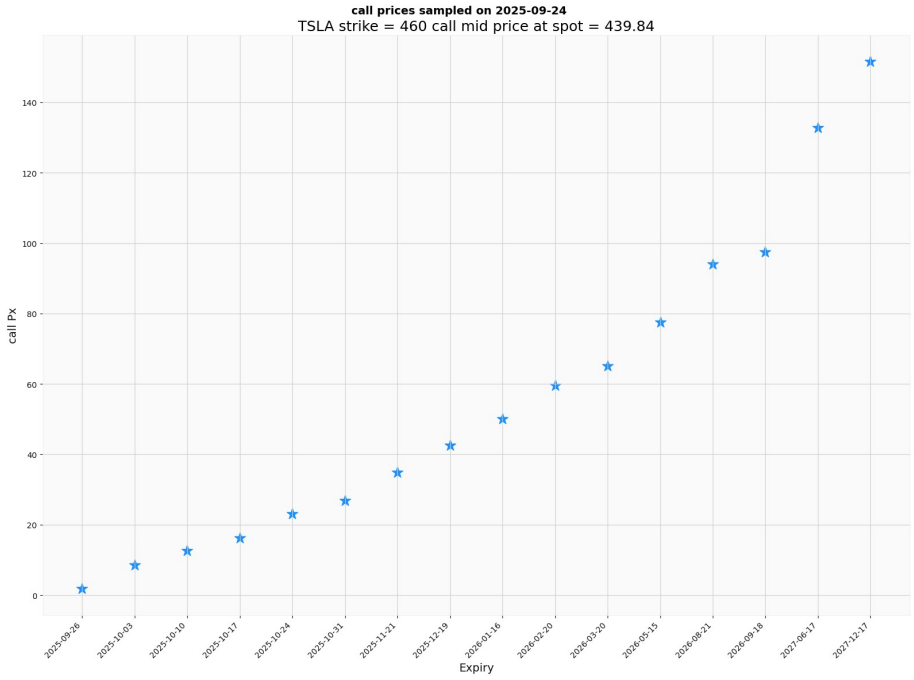


$K_1 < S < K_2$ → strangle < straddle → cheaper to set up

- Calendar spread – same K, different T: let $T_1 < T_2$

  Short T_1 call, long T_2 call:

  - bet on relative value (volatility term structure)
  - use short position to partially fund the long one



(Source: Yahoo Finance)

We will revisit these after we have introduced volatility surface. They are also (simple) examples of manufacturing complex derivatives out of vanilla ones.